

CHAROTAR UNIVERSITY OF SCIENCE & TECHNOLOGY

Seventh Semester of B. Tech. Examination (EE)

December 2012

EE 401 Electrical Machine Design I

Date: 03.12.2012, Monday

Time: 10:00 a.m. To 01:00 p.m.

Maximum Marks: 70

Instructions:

1. The question paper comprises of two sections.
2. Section I and II must be attempted in separate answer sheets.
3. Make suitable assumptions and draw neat figures wherever required.
4. Use of scientific calculator is allowed.

SECTION - I

- Q - 1 (a) For a transformer show that emf per turn, E_t is given by $E_t = k\sqrt{\text{KVA}}$. What are the factors on which the value of K depends? [04]
- (b) A 22 kV, 50 Hz transformer has $I^2 R$, hysteresis and eddy current losses 1.8 %, 1.2 % and 1.6 % of the output. What will be the % losses if the transformer is connected to 11kV, 25 Hz supply assuming the full load current to remain the same? [03]
- Q - 2 (a) Show how output, losses, cost & cost per KVA varies with linear dimensions. [06]
- (b) A three phase, 50 Hz oil cooled, core type transformer has the following dimensions: distance between core centers 0.3 m, height of window 0.42 m, diameter of circumscribing circle 0.15 m, the flux density in the core is 1.55 Tesla and current density in the conductors is 2.2 A/mm². Estimate the KVA rating. Assume a window space factor of 0.29. The core is 4 stepped. [08]

OR

- Q - 2 (a) Derive an expression for KVA rating of a three phase core type transformer in terms of its main dimensions. Also derive an output equation for single phase shell type transformer. [06]
- (b) A 220/110 V, 1 KVA, 50 c/s, single phase transformer has a core with a uniform sectional area of 3700 mm², an effective magnetic core length of 0.6 m and a core weight of 8 kg. If the core is worked at a maximum flux density, B , of 1.7 wb/m², the corresponding magnetizing force, H , is 250 A/m and the specific core loss is 1.0 w/kg, determine (a) the transformer no load current when the hv is fed at 220 V and (b) the corresponding magnetizing reactance and the equivalent shunt resistance to represent the core loss. [08]
- Q - 3 (a) Draw & Describe the constructional detail of bar and wound type current transformer. [04]
- (b) Design and develop a mush winding for a 4 pole, 36 slots, 3 phase stator of AC machine. Develop winding diagram for R phase also show starting of other two phases. [10]

OR

- Q - 3 (a) Explain the use of dummy coils and equalizer connections in dc armature winding. [04]
- (b) Find the front pitch, back pitch, winding pitch & commutator pitch for a simplex wave wound 13 slots, 4 pole dc armature with 13 commutator segments. Draw the [10]

winding diagram in developed form. Also draw sequence diagram to indicate the position of brushes. Assume number of coil sides per slot = 2.

SECTION – II

- Q - 4 (a) Explain the factors to be considered while selecting the number of armature slots in case of D.C. machine. [04]
- (b) Calculate the reactance voltage induced per coil for a single turn two layer winding with two conductors per slot of a 250 kW, 525 V, 6 pole lap wound dc generator driven at 220 rpm. The number of armature conductors is 600. The inductance per coil is 0.0057 mH. The brush covers one commutator segment. If the armature diameter is 1.6 m and core length is 0.3 m, determine the flux density under the inter pole. The length of inter pole is 0.18m. [03]
- Q - 5 (a) Write down the detail design procedure for D.C. shunt and series field windings. [07]
- (b) Find the maximum permissible conductor area for a 1000 kW, 500 V, 8-poles dc generator if the permissible copper loss is not to exceed 300 W/m² of armature surface. The armature diameter is 2.1 m and there are 760 lap connected armature conductors. Assume value of resistivity for copper conductors to be $2 \times 10^{-8} \Omega\text{m}$. [07]

OR

- Q - 5 (a) Discuss the points to be considered while choosing the length of air gap in case of D.C. machine. Find an expression for estimation of air gap length. [07]
- (b) Calculate the following design data for 37kW, 220V, 4 pole, 900 rpm dc shunt motor: [07]
 (i) Length and diameter of core. (ii) Number of armature conductors and type of winding. (iii) Armature line currents. Assume: Specific magnetic loading 0.7 wb/m²; specific electric loading 26000 ampere conductors per metre; ratio pole arc to pole pitch 0.7; length of machine = pole arc; internal drop in armature circuit 10V; field current 2.5 A; field mmf = 1.25 * armature mmf; mmf required for air gap = 0.5 * armature mmf; gap contraction factor 1.15; full load efficiency 89%.
- Q - 6 (a) Derive the equation of temperature rise with time in electric machine. What is heating time constant? [07]
- (b) A motor during colliery winder has to deliver a load rising uniformly from zero to maximum of 1500 kW in 20 second during the accelerating period, 750 kW for 40 second during the full speed period and during the deceleration period of 10 second, when the regenerative braking takes place the kW which is returned to the supply falls from an initial value of 250 to zero. The interval for decking (period of rest) before the next load cycle starts is 20 second. Estimate a suitable rating for the motor. [07]

OR

- Q - 6 (a) Explain the Newton's law of cooling. [07]
- (b) The winding of an oil immersed transformer is 20 mm thick radially. It is wound on a former of negligible thermal conductivity. The power loss in the winding is 25 W/kg, the space factor is 0.75 and the thermal resistivity of the insulating material is 7.5Ω/m. Find the temperature of hot spot above that of winding surface. Copper weighs 8900 kg/m³. [07]
